Solving Linear Inequalities

Exam code: 0607

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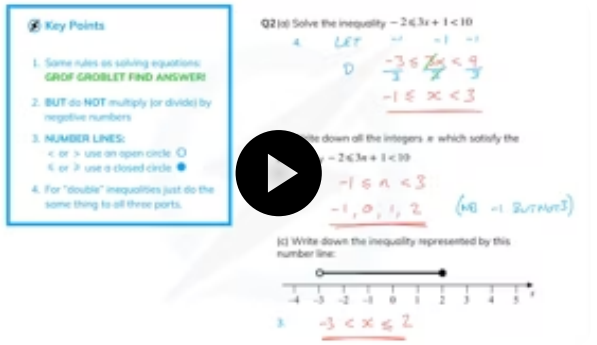

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Solving linear inequalities

What is an inequality?

- An **inequality** tells you that something is **greater than** (>) or **less than** (<) something else
  - $x > 5$ means x is greater than 5
    - x could be 6, 7, 8, 9, ...
- Inequalities may also include being **equal** (=)
  - ⩾ means **greater than or equal to**
  - ⩽ means **less than or equal to**
    - $x \leqslant 10$ means x is less than or equal to 10
      - x could be 10, 9, 8, 7, 6,....
- When they **cannot** be **equal**, they are called **strict** inequalities
  - > and < are **strict** inequalities
    - $x > 5$ **does not** include 5 (strict)
    - $x \geqslant 5$ **does** include 5 (not strict)

How do I find integers that satisfy inequalities?

- You may be given **two end points** and have to list the **integer** values of x that **satisfy** the inequality
  - Look at whether each end point is **included** or not
    - $3 \leqslant x \leqslant 6$
      - $x = 3, 4, 5, 6$

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

        - $3 < x \leq 6$
            - $x = 4, 5, 6$
        - $3 < x < 6$
            - $x = 4, 5$
- If only **one end point** is given, there are an infinite number of integers
    - $x > 2$
        - $x = 3, 4, 5, 6, \ldots$
    - $x \leq 2$
        - $x = 2, 1, 0, -1, -2, \ldots$
        - Remember **zero** and **negative** whole numbers are **integers**
        - If the question had said **positive** integers only then just list $x = 2, 1$
- You may be asked to find integers that satisfy **two** inequalities
    - $0 < x < 5$ and $x \geq 3$
        - List **separately**: $x = 1, 2, 3, 4$ and $x = 3, 4, 5, 6, \ldots$
        - Find the values that appear in **both** lists: $x = 3, 4$
- If the question does not say x is an integer, do **not assume** x is an integer!
    - $x > 3$ actually means **any value** greater than 3
        - 3.1 is possible
        - $\pi = 3.14159\ldots$ is possible
- You may be asked to find the **smallest** or **largest** integer
    - The smallest integer that satisfies $x > 6.5$ is 7

Worked Example

List all the integer values of x that satisfy

$$-4 \leq x < 2$$

Answer:

Integer values are whole numbers
$-4 \leq x$ shows that x includes -4, so this is the first integer

$$x = -4$$

$x < 2$ shows that x does not include 2
Therefore the last integer is $x = 1$

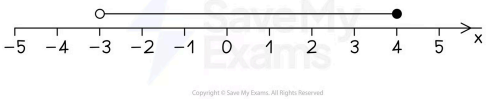
$$x = 1$$

For the answer, list all the integers from -4 to 1
Remember integers can be zero and negative

$$x = -4, -3, -2, -1, 0, 1$$

a number line?

- The inequality $-3 < x \leq 4$ is shown on a **number line** below



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- Draw **circles** above the **end points** and connect them with a **horizontal line**
  - Leave an **open circle** for end points with **strict** inequalities, **< or >**
    - These end points **are not** included
  - Fill in a **solid circle** for end points with **≤ or ≥** inequalities
    - These end points **are** included

< or > use an open circle ○ (end points are excluded)
⩽ or ⩾ use a closed circle ● (end points are included)

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- Use a **horizontal arrow** for inequalities with **one end point**
  - $x > 5$ is an open circle at 5 with a horizontal arrow pointing to the right

Worked Example

Represent the following inequalities on a number line.

(a) $-2 \leq x < 1$

Answer:

-2 is included so use a closed circle

1 is not included so use an open circle

(b) $t < 3$

Answer:

3 is not included so use an open circle

There is no second end point
Any value less than three is accepted, so draw a horizontal arrow to the left

How do I solve linear inequalities?

- Solving linear inequalities is just like **Solving Linear Equations**
  - Follow the same rules, but keep the **inequality sign** throughout
  - If you change the inequality sign to an equals sign you are changing the meaning of the problem
- When you **multiply** or **divide** both sides by a **negative** number, you must **flip the sign** of the inequality
  - E.g.

$$1 < 2$$
$$(\times -1) \qquad (\times -1)$$
$$-1 > -2$$

- **Never multiply** or **divide** by a **variable** (x) as this could be **positive or negative**
- The **safest way** to rearrange is simply to **add** and **subtract** to move all the terms onto **one side**

How do I solve double inequalities?

- Inequalities such as $a < 2x < b$ can be solved by applying the **same operation** to **all three sides** of the inequality
  - E.g. $\frac{a}{2} < x < \frac{b}{2}$
- Alternatively, you can **separate** the inequality into two inequalities
  - $a < 2x$ and $2x < b$

Examiner Tips and Tricks

- Do **not** change the inequality sign to an equals when solving linear inequalities.
  - In an exam you will lose marks for doing this.
- Remember to **reverse the direction** of the inequality sign when multiplying or dividing by a **negative** number!

Worked Example

Solve the inequality $2x - 5 \leq 21$.

Answer:

Add 5 to both sides

$$2x \leq 26$$

Worked Example

Solve the inequality $10 - 2x > 4x - 7$.

Answer:

Add $2x$ to both sides

$$10 > 6x - 7$$

Add 7 to both sides

$$17 > 6x$$

Divide both sides by 6

$$\frac{17}{6} > x \text{ or } x < \frac{17}{6}$$

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